

Universidad Rafael Landivar

Facultad de Ingeniería

Ingeniería en Informática y Sistemas

Redes y telecomunicaciones - sección 1

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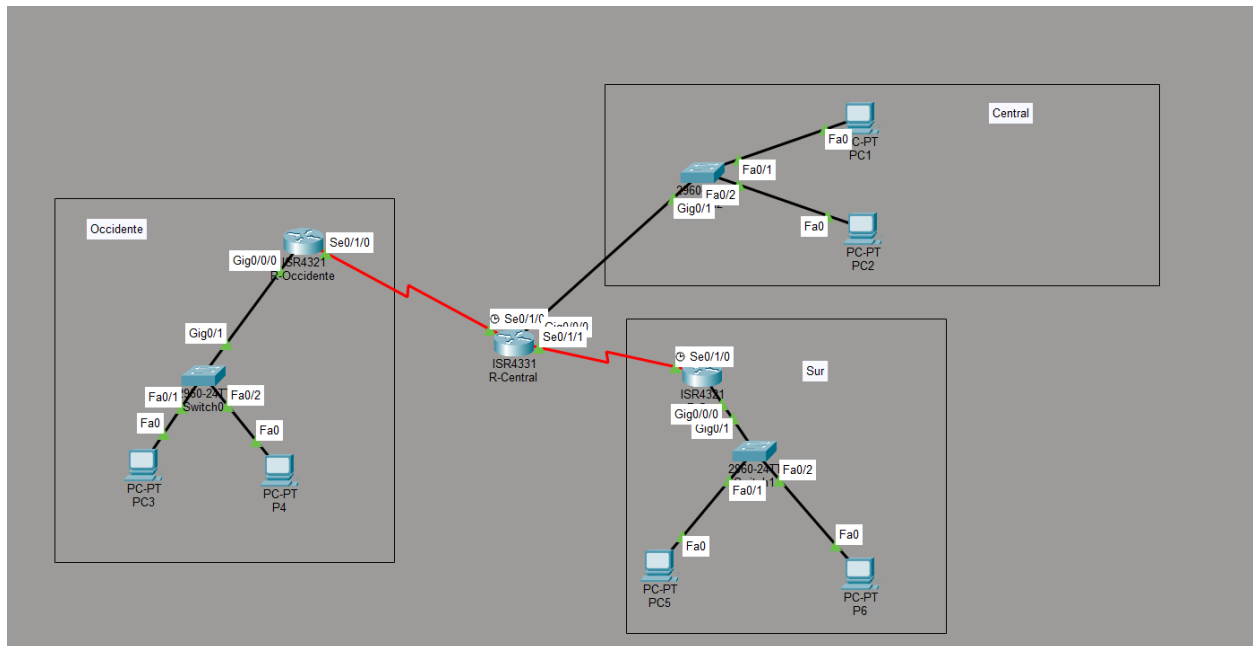
REDES Y TELECOMUNICACIONES:

LABORATORIO 6

Rodrigo José Ruiz Juez: 1037623

Guatemala, 21 de febrero de 2026

Topología de red utilizada



Objetivo del laboratorio

Diseñar e implementar un esquema de direccionamiento IPv4 utilizando VLSM a partir de la red 10.0.0.0/8, configurando enrutamiento estático entre tres campus interconectados mediante enlaces WAN seriales.

Diseño de VLSM

Subred	Host Requeridos	Host Dispon	Dirección de red	Mascara CIDR	Mascara (Decim)	Rango de IP usables	Broadcast
Campus Central	40000	65536	10.0.0.0	/16	255.255.0.0	10.0.0.1 - 10.0.255.254	10.0.255.255
Campus Occidental	20000	32768	10.1.0.0	/17	255.255.128.0	10.1.0.1 - 10.1.127.254	10.1.127.255
Campus Sur	10000	16384	10.1.128.0	/18	255.255.192.0	10.1.128.1 - 10.1.191.254	10.1.191.255
WAN 1	2	2	10.1.192.0	/30	255.255.255.252	10.1.192.1 - 10.1.192.2	10.1.192.3
WAN 2	2	2	10.1.192.4	/30	255.255.255.252	10.1.192.5 - 10.1.192.6	10.1.192.7

configuración de interfaces

R-Central

```
Router>enable
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0/0	10.0.0.1	YES	manual	up	up
GigabitEthernet0/0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/0/2	unassigned	YES	unset	administratively down	down
Serial0/1/0	10.1.192.1	YES	manual	up	up
Serial0/1/1	10.1.192.5	YES	manual	up	up
Serial0/2/0	unassigned	YES	unset	administratively down	down
Serial0/2/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```
Router#
```

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R-Occidente

```
Router>enable
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0/0	10.1.0.1	YES	manual	up	up
GigabitEthernet0/0/1	unassigned	YES	unset	administratively down	down
Serial0/1/0	10.1.192.2	YES	manual	up	up
Serial0/1/1	unassigned	YES	unset	administratively down	down
Serial0/2/0	unassigned	YES	unset	administratively down	down
Serial0/2/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```
Router#
```

R-Sur

```
Router>enable
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0/0	10.1.128.1	YES	manual	up	up
GigabitEthernet0/0/1	unassigned	YES	unset	administratively down	down
Serial0/1/0	10.1.192.6	YES	manual	up	up
Serial0/1/1	unassigned	YES	manual	administratively down	down
Serial0/2/0	unassigned	YES	unset	administratively down	down
Serial0/2/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```
Router#
```

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Enrutamiento

R-Central

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 8 subnets, 5 masks
C       10.0.0.0/16 is directly connected, GigabitEthernet0/0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0/0
S       10.1.0.0/17 [1/0] via 10.1.192.2
S       10.1.128.0/18 [1/0] via 10.1.192.6
C       10.1.192.0/30 is directly connected, Serial0/1/0
L       10.1.192.1/32 is directly connected, Serial0/1/0
C       10.1.192.4/30 is directly connected, Serial0/1/1
L       10.1.192.5/32 is directly connected, Serial0/1/1

Router#
```

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R-Occidental

```
Router>enable
Router#show ip interface brief
Interface          IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0/0 10.1.0.1        YES manual up            up
GigabitEthernet0/0/1 unassigned      YES unset  administratively down down
Serial0/1/0         10.1.192.2      YES manual up            up
Serial0/1/1         unassigned      YES unset  administratively down down
Serial0/2/0         unassigned      YES unset  administratively down down
Serial0/2/1         unassigned      YES unset  administratively down down
Vlan1               unassigned      YES unset  administratively down down

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.1.192.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C       10.1.0.0/17 is directly connected, GigabitEthernet0/0/0
L       10.1.0.1/32 is directly connected, GigabitEthernet0/0/0
C       10.1.192.0/30 is directly connected, Serial0/1/0
L       10.1.192.2/32 is directly connected, Serial0/1/0
S*    0.0.0.0/0 [1/0] via 10.1.192.1

Router#
```

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R-Sur

```
Router>enable
Router#show ip interface brief
Interface      IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0/0  10.1.128.1      YES manual up            up
GigabitEthernet0/0/1  unassigned      YES unset  administratively down down
Serial0/1/0        10.1.192.6      YES manual up            up
Serial0/1/1        unassigned      YES manual administratively down down
Serial0/2/0        unassigned      YES unset  administratively down down
Serial0/2/1        unassigned      YES unset  administratively down down
Vlan1             unassigned      YES unset  administratively down down
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.1.192.5 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C       10.1.128.0/18 is directly connected, GigabitEthernet0/0/0
L       10.1.128.1/32 is directly connected, GigabitEthernet0/0/0
C       10.1.192.4/30 is directly connected, Serial0/1/0
L       10.1.192.6/32 is directly connected, Serial0/1/0
S*    0.0.0.0/0 [1/0] via 10.1.192.5
Router#
```

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Pruebas desde el lado sur al lado central

```
C:\>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```



Top

Pruebas desde el lado sur al lado occidente

```
C:\>ping 10.1.0.10

Pinging 10.1.0.10 with 32 bytes of data:

Reply from 10.1.0.10: bytes=32 time=2ms TTL=125
Reply from 10.1.0.10: bytes=32 time=16ms TTL=125
Reply from 10.1.0.10: bytes=32 time=15ms TTL=125
Reply from 10.1.0.10: bytes=32 time=2ms TTL=125

Ping statistics for 10.1.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 16ms, Average = 8ms

C:\>
```

 [Top](#)

Desde campus central a campus sur

```
C:\>ping 10.1.128.10

Pinging 10.1.128.10 with 32 bytes of data:

Reply from 10.1.128.10: bytes=32 time=1ms TTL=126
Reply from 10.1.128.10: bytes=32 time=8ms TTL=126
Reply from 10.1.128.10: bytes=32 time=1ms TTL=126
Reply from 10.1.128.10: bytes=32 time=1ms TTL=126

Ping statistics for 10.1.128.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 8ms, Average = 2ms

C:\>
```